

Mid-Term Research Report

Agentic AI Customer 360

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1. Preliminary Approach

1. **Event-driven ingestion:** Transaction, Support, Application Usage and KYC events enter a live event stream. Time-based triggers are handled separately for scheduled analysis such as daily rollups and periodic checks.
2. **Parallel specialist agents:** Usage, Support, Transaction and KYC agents independently analyse their respective signals using specialised tools and data sources.
3. **Shared customer state:** Each specialist writes structured findings to a *Per-Customer State Board*. This provides a common state without requiring direct communication between every pair of agents.
4. **Synthesis and life-event inference:** A Synthesis/Correlation Agent combines the specialist findings. The Life-Event Inference stage then identifies higher-level customer situations from the combined evidence.
5. **Conflict resolution:** If the system detects meaningful disagreement or ambiguity, Agent Debate and Resolution is activated. Otherwise, the system continues directly with the resolved customer state.
6. **Decision and action:** The resolved state is passed to Offer/Eligibility and Retention/Action stages. The system produces either a bounded action or an explicit no-action decision.
7. **Refinement and human approval:** A proposed action is checked by the Critique/Compliance-Refiner before reaching a Human-in-the-Loop checkpoint. The human can approve, modify or reject the proposed action before execution.

2. Research Log

2.1 Memory and Context Management

MemGPT: Towards LLMs as Operating Systems

Charles Packer et al., 2023 - <https://arxiv.org/abs/2310.08560>

What the source says: MemGPT studies virtual context management using multiple memory levels so that information outside the active context can still be accessed when needed.

What I used: I used the idea of keeping the active context focused instead of putting the complete customer history into every agent's prompt. Relevant historical information can be retrieved when required.

Architecture decision: The architecture uses Working Memory, Episodic Memory (Vector DB) and Semantic Memory. This three-part separation is my adaptation to the Customer 360 problem (not a direct proposed by MemGPT.)

2.2 Conflict Resolution

Improving Factuality and Reasoning in Language Models through Multiagent Debate

Yilun Du et al., ICML 2024 - <https://arxiv.org/abs/2305.14325>

What the source says: The paper studies multiple language-model instances that propose and debate answers over multiple rounds before producing a final answer. It reports improvements on several reasoning and factuality tasks.

What I used: I used debate in way to explicitly examine competing conclusions when the system has conflicting evidence.

Architecture decision: Agent Debate and Resolution is placed behind a conflict/ambiguity check. Therefore, debate is conditional rather than being run for every customer. The conditional trigger is my design choice; the debate mechanism is motivated by the paper.

2.3 Parallel, Sequential and Refinement Workflows

Building Effective Agents

Anthropic, 2024 - <https://www.anthropic.com/engineering/building-effective-agents>

What the source says: The blog describes composable patterns such as parallelization, sequential workflows and evaluator-optimizer workflows. Parallel processing is useful for independent subtasks, while evaluator- optimizer workflows use evaluation and refinement to improve an output.

What I used: I used parallelization for independent customer-signal analysis and sequential workflows when a later stage requires the output of an earlier stage. The evaluator-refinement idea is used before action execution.

Architecture decision: Usage, Support, Transaction and KYC agents operate in parallel. The main reasoning flow then proceeds through Synthesis, Life-Event Inference, Offer/Eligibility, Retention/Action and Critique/Compliance-Refiner.

2.4 Shared-State Coordination

Exploring Advanced LLM Multi-Agent Systems Based on Blackboard Architecture

Bochen Han and Songmao Zhang, 2025 - <https://arxiv.org/abs/2507.01701>

What the source says: The paper explores a blackboard architecture in which agents with different roles share information through a common blackboard.

What I used: I used the shared-state idea to avoid all-to-all communication structure btw the specialist agents.

Architecture decision: Usage, Support, Transaction and KYC agents write structured findings to a *Per-Customer State Board*. Downstream agents read the relevant customer state from this shared board.

2.5 Human-in-the-Loop

Implement human-in-the-loop confirmation with Amazon Bedrock Agents

<https://aws.amazon.com/blogs/machine-learning/implement-human-in-the-loop-confirmation-with-amazon-bedrock-agents/>

What the source says: The guide describes pausing an agent workflow before a sensitive action so that a human can validate it. It also describes allowing human input to modify parameters or provide additional context.

What I used: I used the principle of inserting human validation before a consequential customer action.

Architecture decision: The proposed action passes through Critique/Compliance-Refiner and then reaches HITL. The human can **Approve**, **Modify** or **Reject**. A modification returns the action to the refinement flow.

2.6 Choosing Multi-Agent Patterns

Choosing the Right Multi-Agent Architecture

Sydney Runkle, LangChain, 2026 - <https://www.langchain.com/blog/choosing-the-right-multi-agent-architecture>

What the source says: The article describes several multi-agent patterns, including parallel execution, routers and handoffs. It also highlights context management and recommends selecting a pattern according to the actual requirements of the task.

What I used: I used this pattern-based view to avoid forcing one coordination method onto the complete system. Different parts of the Customer 360 workflow require different coordination behaviour.

Architecture decision: The architecture uses parallel specialist analysis, sequential handoffs for dependent reasoning, and conditional debate for conflicting conclusions. A shared state board is used to keep specialist findings accessible to downstream stages.

3. Summary of Research-to-Architecture Decisions

Source	Idea Used	Architecture Decision
MemGPT	Managed / selective context	Working, Episodic and Semantic Memory with selective retrieval
Multi-Agent Debate	Compare competing conclusions	Conditional Agent Debate + Resolution
Anthropic	Parallel, sequential and refinement workflows	Parallel specialists + sequential reasoning + action refinement
Blackboard Architecture	Shared state between specialised agents	Per-Customer State Board
AWS HITL	Human validation before consequential action	Approve / Modify / Reject checkpoint before execution
LangChain	Match coordination pattern to task requirements	Combination of parallel, handoff and routing-style coordination

The architecture combines parallel processing for independent customer signals with sequential handoffs when later stages depend on earlier results. Event and scheduled triggers are asynchronous, while reasoning handoffs and human approval are synchronous. Overall, different coordination patterns are used where they fit best: parallel processing for independent signals, sequential handoffs for dependent reasoning, and debate only when there is ambiguity or disagreement.

AGENTIC AI 360

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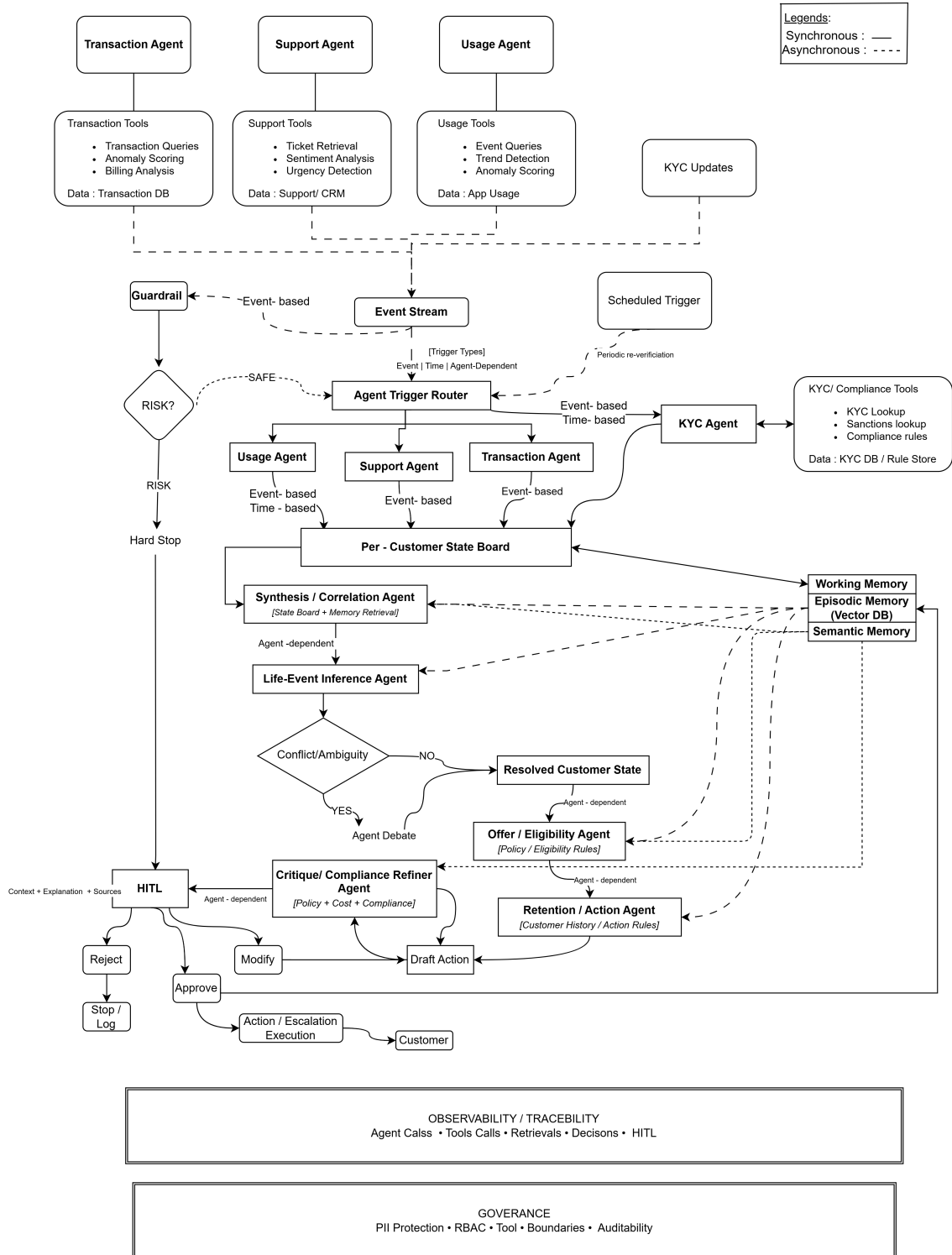


Figure 1: Preliminary Agentic AI Customer 360 architecture.